XRPL Validator — Daily Engineering Report · 2026-10-09

Period: 2026-10-09 00:00 to 24:00 US Eastern (2026-10-09 04:00 to 2026-10-10 04:00 UTC); ledgers #107,530,920 to #107,553,311. Every count from the ledger and the validator's logs stops at #107,553,311, closed 03:59:51 UTC. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,383 of the 22,392 ledgers the network closed (99.960%); every validation matched the network's validated ledger. Unsigned: 9 during planned restarts.
- **Native engine:** 148 engine receipts recorded (105 fixed, 43 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 167), 8 commits merged into main, 1 engine finding fixed (F417).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,392
Validations issued	22,383
Coverage (validations / ledgers closed)	99.960%
Validations matching the network's validated ledger hash	22,383 of 22,383
Unsigned during planned restarts	9
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,383
External agreement, data.xrpl.org (UTC day 2026-10-09)	0.99958 (9 missed of 21,455)

Restarts:

- 16:39:10 UTC: deployment of cycle 167; validations resumed at #107,542,734 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 2,135 transaction-level differences recorded this day, across 641 ledgers. These were not triaged into receipts.

Ledgers with a receipt in the period (#107,530,920–#107,553,311), by step:

Step	Ledgers with a receipt
Step 1 · native engine	148
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	105

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	14:48 UTC	#107,541,000	Step 1 · native engine	1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-02	14:49 UTC	#107,541,023	Step 1 · native engine, Writer	6 transaction result(s) differed from rippled's; 13 change(s) of ours the network did not make; 7 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-03	14:50 UTC	#107,541,029	Step 1 · native engine, Writer	12 transaction result(s) differed from rippled's; 20 change(s) of ours the network did not make; 8 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-04	14:50 UTC	#107,541,030	Step 1 · native	6 transaction result(s) differed from rippled's; 1 ledger change(s) the network made were absent from ours:	fixed: F417 (a Batch inner filed under its ParentBatchID's outer),

ID	Time	Ledger(s)	Step	Description	Disposition
			engine, Writer	51CF5327; 9 change(s) of ours the network did not make; 7 object(s) differed in content	live since 10-09 (cycle 167)
R-05	14:50 UTC	#107,541,031	Step 1 · native engine, Writer	6 transaction result(s) differed from rippled's; 15 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-06	14:50 UTC	#107,541,033	Step 1 · native engine	1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-07	14:51 UTC	#107,541,040	Step 1 · native engine	13 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-08	14:52 UTC	#107,541,061	Step 1 · native engine	6 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-09	14:52 UTC	#107,541,065	Step 1 · native engine, Writer	6 transaction result(s) differed from rippled's; 9 change(s) of ours the network did not make; 9 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-10	14:53 UTC	#107,541,083	Step 1 · native engine	18 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-11	14:54 UTC	#107,541,101	Step 1 · native engine	10 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-12	14:55 UTC	#107,541,103	Step 1 · native engine	3 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-13	14:55 UTC	#107,541,105	Step 1 · native engine	6 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-14	14:55 UTC	#107,541,106	Step 1 · native engine, Writer	6 transaction result(s) differed from rippled's; 15 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-15	14:55 UTC	#107,541,111	Step 1 · native engine	4 change(s) of ours the network did not make; 4 object(s) differed in content	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-16	14:55 UTC	#107,541,117	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-17	14:56 UTC	#107,541,119	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-18	14:56 UTC	#107,541,125	Step 1 · native engine, Writer	12 transaction result(s) differed from rippled's; 19 change(s) of ours the network did not make; 5 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-19	14:56 UTC	#107,541,129	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 4 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-20	14:56 UTC	#107,541,133	Step 1 · native engine	14 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)

ID	Time	Ledger(s)	Step	Description	Disposition
R-21	14:57 UTC	#107,541,143	Step 1 · native engine	5 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-22	14:58 UTC	#107,541,153	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 12 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-23	14:58 UTC	#107,541,155	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 12 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-24	14:58 UTC	#107,541,159	Step 1 · native engine	19 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-25	14:58 UTC	#107,541,161	Step 1 · native engine	3 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-26	14:59 UTC	#107,541,166	Step 1 · native engine	3 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-27	15:00 UTC	#107,541,184	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 12 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-28	15:00 UTC	#107,541,185	Step 1 · native engine, Writer	12 transaction result(s) differed from rippled's; 11 ledger change(s) the network made were absent from ours: EE4B5D51, 1CCB8038, CB151B64; 16 change(s) of ours the network did not make; 24 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-29	15:00 UTC	#107,541,187	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 12 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-30	15:00 UTC	#107,541,188	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 12 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-31	15:00 UTC	#107,541,189	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 4 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-32	15:00 UTC	#107,541,191	Step 1 · native engine	17 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-33	15:01 UTC	#107,541,200	Step 1 · native engine, Writer	6 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: A0835215, D025B37D; 13 change(s) of ours the network did not make; 16 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-34	15:01 UTC	#107,541,202	Step 1 · native engine, Writer	6 transaction result(s) differed from rippled's; 3 ledger change(s) the network made were absent from ours: FF54F33A, A9DC0A65, D7C3A80A; 16 change(s) of ours the network did not make; 7 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-35	15:01 UTC	#107,541,203	Step 1 · native engine, Writer	12 transaction result(s) differed from rippled's; 21 change(s) of ours the network did not make; 20 object(s) differed in content	fixed: F417 (a Batch inner filed under its ParentBatchID's outer), live since 10-09 (cycle 167)
R-36	15:01 UTC	#107,541,204	Step 1 · native engine	1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-37	15:01 UTC	#107,541,206	Step 1 · native engine	1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)

ID	Time	Ledger(s)	Step	Description	Disposition
R-38	15:01 UTC	#107,541,209	Step 1 · native engine	1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-39	15:01 UTC	#107,541,211	Step 1 · native engine	1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-40	15:02 UTC	#107,541,212	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-09-01, INC-01 is INC-2026-10-09-01.

108 more receipts (#107,541,213–#107,542,690) are in the record and not listed individually; their dispositions: 14 explained, 94 fixed.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
167	`b29d83a`	16:39:59 UTC	1,962 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	411 matched, 0 mismatched	#42 from 17:06 UTC: ended early: superseded by the next deployment (2026-10-10 04:45 UTC)

Cycle 167 delivered: `5b6c81f` feat(scripts): watch_engine says who writes state.rocks under the native writer; `d354cf0` feat(tools): xrpl-watch, the validator's terminal dashboard in Rust (HALCYON); `45ca52a` feat(tools): xrpl-watch shows every transaction type and result code; `e53ce35` feat(tools): xrpl-watch lists every type and result on the main screen; `b945ba2` feat(tools): xrpl-watch in real time — ~30 fps, the validator read 4x a second; `c86e5e0` feat(tools): xrpl-watch --record --fps, and a real-time footer; `b29d83a` fix(node): F417 — a Batch inner is filed under the outer its ParentBatchID names (Mainnet #107541023 FAD38FDE4D09)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
16:12 UTC	xrpl-validator	`b29d83a`	fix(node): F417 — a Batch inner is filed under the outer its ParentBatchID names (Mainnet #107541023 FAD38FDE4D09)

Merged	Repo	Commit	Subject
17:10 UTC	xrpl-validator	`59a0ddc`	fix(probe): differential_probe keeps a tecEXPIRED's erasures, as the node does (devnet #5533675 7E7F5D06640B)
22:46 UTC	xrpl-validator	`66081c8`	docs: the plan to a fully independent validator, updated for Stage 4 Phase B
22:51 UTC	xrpl-validator	`7cf3aff`	docs: the independence plan, in depth
23:18 UTC	xrpl-validator	`74a981e`	docs: the independence task breakdown and the recommended order
03:57 UTC	xrpl-validator	`863651f`	fix(ffi): leg A credits a shared Batch inner to the outer its ParentBatchID names, not the last carrier
03:57 UTC	xrpl-validator	`3a9c669`	fix(flow): rippled 3.4.1's overflow hardening — checked savedIns/savedOuts sums in flow() and BookStep
03:57 UTC	xrpl-validator	`5710fd4`	fix(ledger): fixBatchV1_2 — a Batch inner must be a RawTransaction object (rippled 3.4.1)

Findings fixed

- **fix(node): F417 — a Batch inner is filed under the outer its ParentBatchID names (Mainnet #107541023 FAD38FDE4D09)** (`b29d83a`, xrpl-validator) The first mainnet Batches after BatchV1_1 (2026-10-09) came from one bot: up to eight tfAllOrNothing outers from different accounts per ledger, every one wrapping the SAME ticketed pair of inner payments. rippled discards the outers whose round trip falls short of DeliverMin and files the pair under the first one that clears it; the rest get tefNO_TICKET. The engine applied every outer exactly so. But the inner …

Other engineering changes

- **fix(probe): differential_probe keeps a tecEXPIRED's erasures, as the node does (devnet #5533675 7E7F5D06640B)** (`59a0ddc`, xrpl-validator) The probe applies through its own copy of native_apply_one, and that copy restored the post-common snapshot on every claimed tec but tecKILLED. The node's copy settles a tecEXPIRED instead (finding 245: apply::settle_expired keeps the NFTokenOffer / Credential erasures and their directory and OwnerCount moves). So a Payment naming an expired credential (campaign 23, devnet #5533675) kept the credential the network …
- **fix(ffi): leg A credits a shared Batch inner to the outer its ParentBatchID names, not the last carrier** (`863651f`, xrpl-validator) F417's other half. Several outers can carry the very same inner and only one files it (mainnet #107541023: four outers wrapping one ticketed pair, the first discarded, the third filing the pair, the fourth after it). The native side pairs inner verdicts by ParentBatchID since b29d83a; leg A (ffi_engine's C++ compare) still filled `inner_of` in ledger order, so a shared inner's expected mutations folded into the LAST …
- **fix(flow): rippled 3.4.1's overflow hardening — checked savedIns/savedOuts sums in flow() and BookStep** (`3a9c669`, xrpl-validator) rippled 3.4.1 (unconditional, no amendment) folds every strand-level and book-level aggregate through checkedStepAddOpt: XRP (and MPT) add in checked int64, IOU through Number as before.
- **fix(ledger): fixBatchV1_2 — a Batch inner must be a RawTransaction object (rippled 3.4.1)** (`5710fd4`, xrpl-validator) rippled 3.4.1's one gated change (Batch.cpp:293): with fixBatchV1_2, every RawTransactions entry must be wrapped as RawTransaction, or the batch is temMALFORMED. The check is FIRST in the per-inner loop, ahead of the uniqueness check. fixBatchV1_2 was ON on mainnet by #107540484 (2026-10-09), ahead of BatchV1_1, so no mainnet ledger ran Batch without it; devnet and testnet did.

Docs and operations

- **docs: the plan to a fully independent validator, updated for Stage 4 Phase B** (`66081c8`, xrpl-validator)
- **docs: the independence plan, in depth** (`7cf3aff`, xrpl-validator)
- **docs: the independence task breakdown and the recommended order** (`74a981e`, xrpl-validator)

Independence program

No independence record for UTC day 2026-10-09; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,392 (#107,530,920–#107,553,311)
Mean close interval	3.86 s
Longest close interval	9 s (before #107,553,304)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- fixCleanup3_4_0 gained majority — ledger 107,536,129, 2026-10-09 09:34 UTC
- fixBatchV1_2 was enabled — ledger 107,540,481, 2026-10-09 14:15 UTC
- BatchV1_1 was enabled — ledger 107,540,993, 2026-10-09 14:47 UTC

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-09 15:26	10-09 16:12	xrpl-validator	`b29d83a`	fix(node): F417 — a Batch inner is filed under the outer its ParentBatchID names (Mainnet #107541023 FAD38FDE4D09)
10-09 16:09	10-09 17:10	xrpl-validator	`59a0ddc`	fix(probe): differential_probe keeps a tecEXPIRED's erasures, as the node does (devnet #5533675 7E7F5D06640B)
10-09 22:46	10-09 22:46	xrpl-validator	`66081c8`	docs: the plan to a fully independent validator, updated for Stage 4 Phase B
10-09 22:51	10-09 22:51	xrpl-validator	`7cf3aff`	docs: the independence plan, in depth
10-09 23:18	10-09 23:18	xrpl-validator	`74a981e`	docs: the independence task breakdown and the recommended order
10-10 02:42	10-10 03:57	xrpl-validator	`863651f`	fix(ffi): leg A credits a shared Batch inner to the outer its ParentBatchID names, not the last carrier
10-10 02:42	10-10 03:57	xrpl-validator	`3a9c669`	fix(flow): rippled 3.4.1's overflow hardening — checked savedIns/savedOuts sums in flow() and BookStep
10-10 02:42	10-10 03:57	xrpl-validator	`5710fd4`	fix(ledger): fixBatchV1_2 — a Batch inner must be a RawTransaction object (rippled 3.4.1)

Times in UTC. A commit appears on the day it was committed to main or merged into it.